Threshold Effects in the Relationship Between Financial Development and Income Inequality

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Abstract

This paper investigates the existence of nonlinearities in the relationship between financial development and income inequality, using the Instrumental Variable Threshold Regression (IVTR) introduced by Caner and Hansen (2004). It is shown, based on cross-sectional data of 60 developed and developing countries, that there is a significant financial development threshold effect in the finance-inequality nexus. Particularly, our findings are consistent with the Greenwood and Jovanovic hypothesis of an inverted U-shaped relationship between banking sector development and inequality. However, there is no evidence for an inverted U-shaped relationship between stock market development and inequality. Besides, our estimates lead to inconclusive results regarding the impact of stock market development on income distribution in low and lower-middle income countries. Our results show that under the lower regime, the weight of the market is very light and its contribution to income distribution does not matter, while under the higher regime market expansion leads to more inequity.

Keywords: Financial development, income inequality, IV threshold regression, panel data
JEL classifications: O16, O15, C36, C33

1. Introduction

During the recent decades, the social and economic inequalities have increased in most of the countries. According to the OECD (2015a), the gap between rich and poor is at his highest level since 30 years. In the 1980s, 10% of the wealthiest households in OECD had an average income level that is seven times higher than the poorest 10% (OECD, 2011). This ratio increased to 8:1 in the 1990s, and 9:1 in the 2000s (OECD, 2015a). In 2015, the average disposable income of the richest 10% in OECD countries was around 9.6 times higher than that

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of the poorest 10%. In some European countries the gap is less noticeable since the ratio is less than 7 to 1 in France and Germany, but it is 18 to 1 in the United States. Income inequality even worsened in egalitarian tradition countries, such as Sweden, where the ratio rose from 3.5 to 1 in 1985 to 6.5 to 1 in 2013. On the other side, income inequality has significantly narrowed in several emerging and developing countries, but the gap between rich and poor remains striking. In Chile and Mexico, the richest 10% of the population earn 25 times the income of the poorest 10%. (OECD, 2015a).

The increase of these inequalities is not only a social or ethical problem, but also, both, a cause and a consequence of economic difficulties. The redistribution policies proved to be insufficient to reduce income inequality. That is, the deficit reduction initiated by governments has led to significant cuts in transfer programs and public services which reduced the redistribution capacity of states and promoted the income gaps growth.

The recent financial crisis, which is mainly the result of financial development, has contributed to the amplification of the phenomenon. The long-run dynamics of rising inequality may itself be the cause of this crisis (Berg and Ostry, 2011). Actually, the global financial crisis has raised deep questions about the role of finance and pushed back the question on the relationship between finance and income inequality,

For economists the challenge would be to provide policymakers with a clear and understandable framework that explains the ways and channels through which financial development may reduce income inequalities. Unfortunately, the existing literature shows conflicting evidence. We believe these mitigated findings are mainly explained by the gap existing between theory and empirical investigations that have addressed the finance-inequality nexus.

In this paper, we argue that, basically, the relationship between financial development and income inequality is nonlinear and subject to threshold effects. Standard regression models

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assuming a linear relationship between the two variables may have led to biased and misleading results. Therefore, we choose to apply the threshold model proposed by Caner and Hansen (2004) to test the presence of these threshold effects, using cross sectional data for 60 developed and developing countries. Our main assumption is that the distributional effect of financial development depends on the degree of development of the financial system itself.

By using a panel threshold model, the present study contributes to the existing literature in four ways. First, this model has the advantage to detect the potential nonlinearity surrounding the relationship between financial development and income inequality. Secondly, by using the Caner and Hansen (2004) IV threshold regression technique, we overcome the potential endogeneity problem via a set of instrumental variables. Third, the panel threshold regression is suitable to deal with the cross-section heterogeneity problem[1]. Finally, this model is a step towards filling the gap in the literature between theory and empirical investigations about the finance/inequality nexus, by combining the two main hypotheses, i.e. the "inequality-widening hypothesis" and the "inequality-narrowing hypothesis".

The remainder of the paper is organized as follows. Section 2 presents a review of the relevant literature, emphasizing nonlinearities in the finance-inequality nexus. Section 3 describes the threshold model to be estimated. Section 4 discusses the data-related issues. Section 5 presents the main empirical findings, while section 6 concludes.

2. Literature review

There is a widespread idea supporting that financial development reduces inequality. Recent developments show that in the presence of investment indivisibilities and increasing returns to scale, the financial imperfections resulting from information asymmetries and transaction costs

[1] Pesaran and Smith (1995) and Hsiao (2003) have already indicated that ignorance of this issue causes biases to appear.

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produce credit constraints that are particularly harmful for small businesses and poor people lacking collateral and connections. By reducing financial frictions and by allowing the poorest people and businesses to access and obtain external financing, financial development can improve the allocation of capital and reduce income inequality (Galor and Zeira, 1993; Aghion and Bolton, 1997; Galor and Moav, 2004).

Galor and Zeira (1993) believe that an economy with capital market imperfections and unequal initial distribution of wealth will maintain this inequality and grow more slowly than a similar economy with more equitable initial distribution of wealth. Similar predictions have been shown by Banerjee and Newman (1993).

However, other studies argue that financial development exacerbates income inequality and helps mainly the rich (Haber et al, 2003; Greenwood and Jovanovic, 1990). Haber et al. (2003) consider that the credit is generally channeled to connected entrepreneurs and not to those with better opportunities and prospects. The poor mainly rely on informal family connections for funds, so that financial depth excessively benefits the rich. Controversially, Stiglitz (2015) argues that the financial sector, through a variety of forms of exploitation rents, has contributed to both the inequalities at the top and the bottom in the US.

In their seminal paper, Greenwood and Jovanovic (1990) studied the link between finance and inequality in the context of an endogenous growth model. They found that the dynamics of the development process are nonlinear and support the Kuznets hypothesis (von Bayern, Heathcote et al. 2009). In the early stages of economic development when financial markets are virtually nonexistent, financial development increases inequality. Then, when the economy strengthens and an extensive financial structure develops, the distribution of income becomes more stable. This form of inverted U-shaped curve is based on fixed costs that arise when people want to resort to financial intermediaries.

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Many empirical studies have addressed the finance-inequality question. But, the results were mixed. Das and Mohapatra (2003) examined how the liberalization of the stock market affects the distribution of income. They conclude that the gains from liberalization appear to be in favor of the top quintile to the cost of middle quintiles and without effect on the lower income quintile.

Liang (2006) studied the link between finance and inequality in urban China. His main finding was that the relationship is linear and the financial development of China contributes significantly to reducing income inequality in urban areas. However, these positive distributional gains from financial sector development have been severely reduced by the increase in urban unemployment and massive layoffs caused by the radical reforms in urban areas and by restructuring of public enterprises.

Clarke et al. (2006) studied the relationship between finance and income inequality based on a sample of 83 developed and developing countries for the period 1960 -1995. They found that, in the long-run, the greater is financial sector developed, the lower is inequality.

Beck et al. (2004, 2007) used a sample of 72 developed and developing countries over the 1960-2005 period. Their results showed that financial development disproportionately boosts incomes of the poorest quintile and reduces inequalities.

Recently, Beck et al. (2009) studied the effect of banking deregulation on income distribution in the United States over the 1976-2006 period. They found that deregulation has significantly enhanced income distribution. The results suggest that regulatory barriers to competition between banks in the 20th century were disproportionately detrimental to low-income workers. In contrast, Gimet and Segot (2011) used a Structural Vector Autoregressive Model and a sample of 49 countries over the period between 1994 and 2002. They found that the banking sector development tends to increase income inequality and that this relationship depends on the characteristics of the financial sector rather than its size.

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Similarly, Jauch and Watzka (2015) used an unbalanced panel of 138 developed and developing countries for the 1960-2008 period to investigate the finance- inequality nexus. Their empirical results reject the theoretical models predicting a negative impact of financial development on income inequality measured by the Gini coefficient.

Another but more recent country-level study by Sehrawat and Giri (2015) examined the finance–inequality nexus in India between 1982 and 2012. The results show that financial development aggravates income inequality in both the long-run and the short-run.

This result is contrasted by Chen and Kinkyo (2016). Their estimation results from a panel error-correction model using the PMG estimator suggest that financial development reduces inequality only in the long-run. However, estimates of country-specific coefficients exhibit adverse short-run effects of financial development on income inequality.

More recently, Seven and Coskun (2016) using dynamic panel data methods to assess the finance– inequality–poverty nexus argued that although financial development promotes economic growth, this does not necessarily benefit those on low-incomes in emerging countries.

Obviously, the results derived from the above studies are not conclusive in matters of the exact relationship between finance and income inequalities. As argued by Altunbaş and Thornton (2019), existing empirical literature examining the financial development/inequality nexus provides inconclusive findings because most of empirical studies are based on ordinary least squares, instrumental variables, and panel estimation techniques, which do not allow to catch the effects of the explanatory variables on the entire conditional distribution of the dependent variable.

As an alternative method, Altunbaş and Thornton (2019) propose a quantile regression approach to examine the effect of financial development on income inequality. Their empirical results suggest the impact of financial development on income inequality changes with a

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country's income level. In particular, financial development is found to promote equality in upper-middle income countries, whereas it accentuates inequalities in low- and high-income countries.

Another possibly reliable explanation that can be conveyed to explain these divergent findings is the presence of threshold effects. Ineluctably, empirical analyses that omit this feature could lead to spurious regressions and probably misleading results.

Actually, only few empirical studies have attempted to test the presence of threshold effects in the relationship between financial development and income inequality. For instance, Bresson (2004) tried to capture the effects of financial development channeled through investment in physical capital and educational capital on income inequality. His study was conducted on a sample of 130 countries over the 1960-1999 period. The results show that the overall effect of financial development on inequality quickly becomes positive above a threshold of a relatively low income. Nevertheless, he found a reducing inequality through the influence of banking sector development on the distribution of education.

Bonfiglioli (2005), using a sample of 69 countries over the 1980-2000 period, showed that a better investor protection contributes to financial development and affects income inequality in different ways. Indeed, any improvement over a low level of investor protection leads to an increase of risk-taking rather than risk-sharing, which leads to an increase in income inequality. However, when investor protection is sufficiently high, further improvement is more favorable for risk-sharing than risk-taking and inequality is reduced.

More recently, Bacarreza and Rioja (2008) studied the effects of financial development on income distribution in the region of Latin America and the Caribbean during the period from 1960 to 2005 using a dynamic panel estimation. Their main finding is that financial development affects disproportionate effects on income quintile groups. Particularly, the

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positive distributional effects of the financial system expansion are not captured by the poorest quintile. The authors also noted some evidence for the Greenwood and Jovanovic hypothesis, in such a manner that the positive effect begins only beyond a given threshold of economic development.

On the other hand, Kim and Lin (2011), using a sample of 53 countries over the 1960-2005 period, proved the Greenwood and Jovanovic hypothesis of nonlinearity. They found that the benefits of the banking sector and the stock market development occur only if the country reaches a threshold level of this development. Below this critical value, financial deepening opposes the income inequality reduction.

Tan and Law (2012) found that in the early stage of financial development, financial deepening leads to more equitable distribution of income, but after reaching a higher level of financial development, excessive financial deepening widens the gap between the rich and the poor.

Similarly, Kunieda et al. (2014) showed that financial development expands income inequality under higher financial openness, whereas it reduces income inequality under lower financial openness. This result is confirmed by Park and Shin (2017) who highlighted a U-shaped relationship between finance and inequality. That is, financial development contributes to reducing inequality up to a point, but as financial development proceeds further, it contributes to greater inequality.

However, these finding are contrasted by the study conducted by Kaidi and Mensi (2016). Based on an international sample of 138 countries over the period 1980-2012, their findings suggest an inverted U-shaped relationship between financial development and income inequality.

Using a sample of 23 Chinese provinces during the period 1996–2012, Liu and all (2016) show, based on a panel threshold regression model, that financial deepening helps to reduce income

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inequality only in the lower financial development regime. This positive relationship turns out to be insignificant if the stock market capitalization has reached a certain threshold. A similar result has been recently reported by Shahbaz et al (2017) who found a U-shaped relationship between financial development and income inequality in Kazakhstan over the period 1990-2014. That is, financial development narrows income inequality in the early stages of transition, but beyond a given threshold it leads to an increase in income inequality.

In the same direction, Baiardi and Morana (2018) detected a "too much finance" effect during the recent crises in the Euro area. Their empirical findings show that income inequality falls as financial development increases up to a threshold value of 90–100 GDP points, and then rises as financial deepening exceeds the threshold.

More recently, Chiu and Lee (2019) explored the non-linear relationships among income inequality and financial development when country risk environments change, by using a panel smooth transition regression model. Their results show the effect of financial development on income equality is positive for high-income, whilst it is negative for low-income countries.

2. The model

We estimate a model similar to that of Beck and Levine (2004), Beck et al. (2007), Ang (2010) and Rioja and al. (2014), where change in income inequality is regressed on financial development and a set of control variables:

$$\Delta inequality_i = \beta F_i + \delta' X_i + \varepsilon_i \qquad (1)$$

where $i = 1, 2 \ldots N$ is the country indicator and ε_i is an error term. $\Delta inequality_i$ denotes changes in income inequality, F_i is the financial development variable and X_i represents a vector of control variables including a constant term.

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To allow for nonlinearity in the relationship, we extend equation (1) into a threshold regression model that takes the following form:

$$y_i = \theta_1' z_i I(q_i \leq \gamma) + \theta_2' z_i I(q_i > \gamma) + \varepsilon_i \qquad (2)$$

where $y_i = \Delta inequality_i$, $z_i = (F_i, X_i')'$, $\theta_j = (\beta, \delta')'$, j=1,2. I(.) represents an indicator function which takes the value 1 if the argument in parenthesis is valid, and 0 otherwise. q_i is an exogenous threshold variable used to split the data into different regimes or groups. The threshold parameter is $\gamma \in \Gamma$, where Γa strict subset of the support of q_i. This model allows the regression parameters θ_1 and $\theta_2$to switch between regimes depending on whether q_i is smaller or larger than the unknown threshold value γ.

Equation (2) can be re-written in a more detailed form as follows:

$$\Delta inequality_i = \beta_1 F_i\, I(q_i \leq \gamma) + \beta_2 F_i\, I(q_i > \gamma) + \delta_1' X_i\, I(q_i \leq \gamma) + \delta_2' X_i\, I(q_i > \gamma) + \varepsilon_i \quad (3)$$

Assuming that all explanatory variables are exogenous, Hansen (2000) suggests equation (2) to be estimated using Ordinary Least Square (OLS). Conditional on the above estimators, we then can estimate the threshold value by minimizing the concentrated sum of squared errors function (SSE).

However, the threshold estimation method proposed by Hansen (1996, 2000) is not suitable when one or more explanatory variables are endogenous. Since we consider that financial development is highly likely to be endogenous, we rather choose to estimate equation (2) by applying the Caner and Hansen (2004) instrumental variable threshold regression model (IVTR).

The authors define a reduced form of the endogenous variable as follows:

$$F_i = g\,(s_i, \pi) + u_i \qquad (3)$$

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where π_i is an unknown parameter vector, g is a linear function and u_i is a random error. The vector s_i represents a set of instrumental variables which are not included in the inequality regression, along with other exogenous variables of the model. After replacing the endogenous variable by its reduced form, equation (3) can be written as follows:

$$\Delta inequalityi = \beta_1\, g\,(s_i, \pi)\, I(q_i \leq \gamma) + \beta_2\, g\,(s_i, \pi)\, I(q_i > \gamma) + \delta_1{}' X_i\, I(q_i \leq \gamma) + \delta_2{}' X_i\, I(q_i > \gamma) + vi \qquad (4)$$

where $vi = \beta_1 u_i\ I\,(q_i \leq \gamma) + \beta_2\, u_i\, I\,(q_i > \gamma) + \varepsilon i$

The parameters are estimated sequentially. First we estimate the reduced form parameter π and obtain the predicted value $\widehat{F_i} = \hat{g}_i = \mathrm{g}\,(\mathrm{s_i}, \hat{\pi}) = \hat{\pi}'\mathrm{s_i}$. Second we turn to the estimation of the threshold parameter γ which is chosen to minimise the sum of squared residuals from a sequence of regressions of inequality variable on the predicted value of financial development variable. Third, we estimate the slope parameters $(\beta_1, \beta_2, \delta_1, \delta_2)$ by 2SLS or GMM on the split sample implied by $\hat{\gamma}$.

To test the existence of threshold effect, Caner and Hansen (2004) suggest the Supremum Wald test statistic. First, we fixe $\gamma \in \Gamma$ to any value and the equation (2) is estimated by GMM. Then, for the fixed value of γ, the Wald statistic under H_0 is constructed as follows:

$$w_n(\gamma) = \left[\hat{\theta}_1(\gamma) - \hat{\theta}_2(\gamma)\right]' \left[\hat{V}_1(\gamma) + \hat{V}_2(\gamma)\right]^{-1} \left[\hat{\theta}_1(\gamma) - \hat{\theta}_2(\gamma)\right] \qquad (5)$$

Repeating the calculation for all $\gamma \in \Gamma$, we can obtain the Sup statistic as the greatest value of the calculated w_n:

$$Sup\ W = \sup_{\gamma \in \tau} w_n(\gamma) \qquad (6)$$

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The asymptotic distribution of this statistic is not standard because of the presence of the nuisance parameter γ. For this reason, Caner and Hansen (2004) use a bootstrapping procedure to get the p-value.

Finally, given the estimate $\hat{\gamma}$ of the threshold variable, the sample is split into two subsamples (regimes) based on the indicators $I(q_i \leq \hat{\gamma})$ and $I(q_i > \hat{\gamma})$. The slope parameters are then estimated separately on each subsample. Caner and Hansen (2004) show that the estimators are consistent and the slope parameter estimates asymptotically normal with conventional covariance matrices.

3. Data description and sources

The sample data used in this study is a pool of 60 developed and developing countries (25 high-income countries, 30 middle-income countries and five low-income countries) over the 1980-2009 period. It includes 17 countries in North America, Europe and Central Asia, 17 in Latin America and Caribbean, 14 in South and East Asia and Pacific, 8 in Sub-Saharan Africa and 4 in Middle East and North Africa[2].

All variables are averaged over the sample period. The income inequality data were obtained from the World development Indicators (WDI). For the financial development indicators, we use the "Financial Structure Database" compiled by Beck et al. (2009). Data sources and definitions are detailed in Table A1 in the Annexes.

3.1 Income inequality variables

Following Beck et al. (2004, 2007) and Kim and Lin (2011), changes in income inequality are captured through the annual growth rate of the Gini Coefficient (GINI_GR), computed as the log difference between the last and the first observation divided by the number of years between

[2] A complete list of countries is provided in table A3.1 in the Annexes.

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these two observations. Nevertheless, since the Gini coefficient gives only information on the overall income distribution, we need another measure to ensure that the change in income inequality is associated with a change in the income of the poor. Consequently, we used the annual growth rate of the share of the lowest Income quintile (LOWEST_GR) as an additional income inequality measure, which is computed as the log difference between the share of the lowest income quintile for the last and the first observations divided by the number of years between these two observations.

3.2. Financial development variables

As advanced by Lynch (1996), a good measurement of financial development is crucial for effective policy formulation, implementation and evaluation. We believe this argument holds particularly true in assessing how financial development reduces poverty or income inequality. But as pointed by Beck and al. (2007), a perfect indicator of financial development does not exist. For this reason, we rely in this study on the most commonly used proxies for financial development. Firstly, to proxy the banking sector development we followed Beck and Levine (2002) and Kim and Lin (2011) using three indicators expressed in real terms. The first one is Private Credit (PRIV) which equals the value of credit by financial intermediaries to the private sector divided by GDP. To detect the performance of banking activities, we used the Liquidity Ratio (LLY). It is measured as the ratio of M2 (broad money) to GDP. This ratio determines the size of financial intermediaries without limitation to the private sector. The third indicator is Bank Assets (DBA), measured by the ratio of domestic deposit money bank assets to GDP. This proxy reflects the degree of saving allocation to companies by domestic banks.

As for the measurement of the stock market development, we have followed Rousseau and Wachtel (2000), Beck and Levine (2002) and Kim and Lin (2011), by using three proxies: i) The Stock Market Capitalization (MCAP) equals the total value of listed shares divided by

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GDP. It reflects the importance of financing by issuing shares in raising capital and in the process of resource allocation. ii) The Turnover Ratio (TOR) is equal to the value of traded shares on domestic exchanges divided by the value of listed shares. This liquidity measure enables detecting the efficiency of the national Stock Market. An increase in TOR implies lower transaction costs. iii) Finally, the Value Traded (VT) is defined as the value of traded shares on domestic exchanges divided by GDP. It indicates the volume of stock market transactions relative to the size of the economy.

3.3. Control variables

In line with empirical literature on growth and finance, we introduce in the regression a set of control variables in accordance with the augmented Solow model as proposed by Mankiw and al. (1992).

The growth rate of real GDP per capita (GDP_GR) is included to capture the Kuznets hypothesis. The average years of the secondary school attainment variable (SCHOOL) is added to control the effect of human capital. Inflation (INFL) is introduced in the regression to account for the macroeconomic environment. The ratio of government expenditure to GDP (GOV) is included to account for the degree of government involvement in markets and the possible use of redistribution expenditure. Trade openness (TRADE) is measured by the sum of exports and imports divided by GDP. It is included in the regression to capture the degree of international openness.

Furthermore, to alleviate the problem of simultaneity and to assess whether the initial conditions of a country matter for future income inequality, all control variables are taken in logarithm and in initial values.

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3.4. Instrumental variables

As suggested in the literature, we include the initial values of financial development indicators and creditor rights (Kim and Lin, 2011) and the levels of religious composition, ethnic fractionalization and legal origins (MacCaig and Stengos, 2005; Beck et al., 2004) as instruments in order to control for endogeneity and reverse causality.

For creditor rights (CREDITOR), the data were taken from Djankov et al. (2005). This measure reflects the legal rights of creditors against defaulting debtors; a higher value indicates stronger creditor rights. The legal origin proxy is a dummy variable (BRITISH) that takes 1 if the legal system is based on British tradition and 0 otherwise. La Porta et al. (1997, 1998) argue that the origin of legal system has consequences on contractual institutions and show that investor protection and the efficiency of the courts differ between different legal origins. Indeed, countries whose legal sources are based on British tradition ("Common Law") offer better protection to creditors and shareholders' than countries with a legal system based on French tradition ("Civil Law"). As regards to the religious composition, Landes (1998) believes that Catholic and Muslim countries are characterized by cultural intolerance that limits competition and property rights protection. In this context, we consider two variables that reflect the part of Catholics (CATHOLIC) and Muslims (MUSLIM) in the population since these two indicators have strong influence on financial development. Finally, for ethnic fractionalisation (ETHNIC), Beck et al. (2003) advanced that greater ethnic diversity implies the adoption of policies and institutions that rely on maintaining power and control rather than the creation of an open and competitive financial system.

INSERT TABLE 1 HERE

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Summary statistics for the dependent and explanatory variables are given in table 1. Besides, as multiple regression analysis is often subject to multicollinearity problems, we constructed a correlation matrix between financial development and inequality measures. Unsurprisingly, Table 2 shows that variables are highly correlated. For this reason, we applied a sequential approach in which regression analyses were conducted separately for each of the financial development and inequality measures.

INSERT TABLE 2.1 HERE

4. Results

Table A.2 reports parameters estimates of the linear GMM model. It is shown that the impact of financial development on income inequality is insignificant regardless of the measure used of financial development. Dollar and Kraay (2002) have already explained this result by the fact that financial development helps the poor and the rich in the same proportions. Kim and Lin (2011) argue that the use of a sample of countries with different levels of financial development may obscure the different roles played by this development in reducing inequalities. Thus, ignorance of the potential threshold effects in the relationship between financial development and inequality may cause biases to appear leading to fallacious results.

4.1. Linearity tests

The linearity tests have been conducted by using the initial values of three banking sector development indicators, i.e. $PRIV_0$, LLY_0 and DBA_0, and three stock market development indicators, i.e. TOR0, VT0 and MCAP0, as potential threshold variables[3]. Furthermore, the

[3] As recommended by Caner and Hansen (2004.), the threshold variable should be exogenous. For this reason, we only consider the initial values of the financial development variables (i.e. $PRIV_0$, LLY_0, DBA_0, $MCAP_0$, TOR_0, VT_0).

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Wald–Wolfowitz test has been applied to check for randomness[4]. Obviously, results shown in table 2.2 indicate that we cannot reject the null hypothesis that the variables are in random order at the 5% significance level.

INSERT TABLE 2.2 HERE

As discussed in section 2, to test whether there is evidence for a threshold effect in the relationship between financial development and income inequality we employ the Sup *W* statistic. The asymptotic p-values are computed by following the bootstrap procedure suggested by Caner and Hansen (2004)[5].

Estimation results reported in tables 3-6 indicate that the Greenwood and Jovanovic (1990) hypothesis of an inverted U-shaped relationship between financial development and inequality cannot be rejected. The results show a strong and significant threshold effect in the financial intermediary development-inequality and stock market development-inequality nexuses. Across alternative measures of financial development, the p-values of the Wald test show that the threshold effects of financial variables on GINI_GR and LOWEST_GR are statistically significant.

4.2. Financial intermediary development and income inequality

The linearity tests show that the estimated threshold varies between 25.09% and 26.43% when the GINI_GR is used as a dependent variable (Table 3) and between 26.90 % and 40.19% for LOWEST_GR inequality measure (Table 4). These threshold values indicate the degree of financial intermediary development a country should achieve so that improvement in its

[4] As suggested by an anonymous referee, linearity tests results would be less convincing if the 6 potential threshold variables were roughly clustered around two modes and not uniformly distributed across the 60 countries.

[5] Linearity tests have been performed and confidence intervals constructed using 10 thousands bootstraps.

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financial system may lead to a reduction of income inequality. Thus, the threshold regression splits the sample into two separate regimes: one is characterized by a developed banking system and the other by a less developed one. As shown in tables A3.2 and A3.3, the high regime includes 32 (21) countries among them 88% (90%) when $PRIV_0$ is used as threshold variable and GINI_GR (LOWEST_GR) is used as a dependent variable. When LLY_0 is taken as threshold variable, 32 countries fall into the high financial regime, among them 76% are upper-middle and high-income countries.

One can realize, based on estimation of the threshold model, that the coefficients on financial development are statistically significant for the two regimes. Particularly, the results presented in Tables 3 and 4 indicate that bank development has a threshold and statistically significant effect on both the growth of the Gini coefficient and the growth of the income share of the poorest quintile. This result implies that during its first stages of development, financial intermediary accentuates inequalities and disproportionately helps more the rich. The positive distributional effects of financial development are not effective unless the banking sector has reached a minimum critical size. In other words, in a system with less developed financial intermediation, expansion of the banking sector raises inequalities and is detrimental to the poor. However, in the developed banking system regime, financial development positively affects income distribution and, more interestingly, boosts the income of the poor. Thus, a better allocation of resources to the private sector and a relaxation of credit constraints might contribute to reducing inequalities in countries with a developed banking system. However, it deteriorates the income distribution in countries where the banking system is underdeveloped. Obviously, the threshold model proves that the relation between financial development and income inequality is nonlinear. Mitigated results from previous studies are likely due to the misleading assumption of a linear and monotonic relationship between the two variables.

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INSERT TABLE 3 HERE

Tables 3 and 4 show another interesting result. In the threshold regression, initial income share of the poor and initial Gini both enter negatively and significantly. This result suggests that countries with initially less equitable distribution of income might experience faster reductions in income inequality. However, these reductions are expected to be more important in those countries with a more developed banking system, as the effect tends to be bigger in the higher regime.

Furthermore, tables 3 and 4 provide interesting insight regarding the behavior of some control variables. Initial schooling enters significantly in the threshold regression. The higher is the initial secondary school enrolment rate, the faster are the reductions in income inequality and the bigger is the share of income received by the poorest quintile. Besides, the coefficient estimates suggest that schooling has a significant negative (positive) effect of on the Gini growth rate (growth of the share of the poorest quintile), regardless of the level of development of the financial system. This result confirms the tight relationship between access to education, economic growth and poverty alleviation.

On the other hand, the relationship between initial trade openness and income inequality is not clear. Under the developed banking system regime, international openness aggravates income inequality and hurts the poor. Nevertheless, the negative distributional effect of international openness is uncertain under a less developed banking system regime. When private credit is considered as a threshold, trade openness is not found to be pro-poor. Trade openness is unlikely to be reducing poverty when the poor do not have access to credit. We also find that the coefficient estimates from table 4 show that high levels of inflation are correlated, positively and significantly, with greater income inequality. This is true independently from the level of the banking system development. The result corroborates the idea that inflation hurts the poor

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by redistributing incomes to the detriment of those already living in precarious conditions and least accessing financial instruments to hedge against price instability. In addition, the results show that government expenditures play a significant role in reducing income inequality. Nevertheless, it seems this role is more important under a less developed financial intermediation regime.

INSERT TABLE 4 HERE

Finally, the threshold regression results show that the initial GDP growth rate has a nonlinear effect on income inequality, depending on the degree of financial intermediary development. Actually, the results fail to provide any support for the Kuznets hypothesis. The final effect of economic growth on income inequality appears to depend ultimately upon the structure of the financial system per se. Our results suggest that countries initially growing rapidly might experience faster reductions in income inequality. However, these reductions are expected to be more effective in countries with a less developed banking system. In fact, large banking sectors tend to pump excessive credit to households, inflating unsustainable bubbles, which disproportionately benefit the rich. This result seems accurately describing what happened in the OECD countries in the aftermath of the global financial crisis (OECD, 2015).

4.3. Stock market development and income inequality

When considering stock market development as threshold variable, results from linearity test show again a significant threshold effect in the financial development–inequality nexus.

INSERT TABLE 5 HERE

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As shown in Table 5, under regimes with less developed stock market, financial development has no significant impact on inequality unless a given financial development level is reached. Surprisingly, above this threshold level stock market development accentuates the income inequality, as measured by the growth of the Gini index.

INSERT TABLE 6 HERE

Estimation of the threshold model with the of lowest income quintile growth as a dependent variable helps to better understand this result. From table 6, it is shown that inequality is not related to a decrease in the income of the poor, but possibly to an increase of the middle class or the rich income. However, such a result should be interpreted with caution. Estimations from table 6 lead to mitigated and inconclusive results as regards the impact of stock market development on the income of the poor. These conflicting results suggest that changes in income inequality may be related to a change in the middle class and the rich's income and not to the poor's one.

Ultimately, the results of the threshold regressions between stock market development indicators and inequality measures suggest a negative and statistically significant effect of stock market development on growth of the average income of the poorest quintile, and positive and statistically significant on growth of the Gini coefficient. But this result is restricted for the high financial regime. Tables A-3.4 and A-3.5 provide insightful information about classification of countries under the two financial regimes. Unsurprisingly, upper-middle and high income countries represent more than 90% of total countries falling into the high regime[6]. Such information indicates the strong correlation between financial development and economic growth.

[6] Upper-middle and high income countries represent, respectively, 86% and 96% of total countries falling into the high regime when MCAP and TOR are used as stock market development indicators under the "growth of Gini" model specification, and 90% when VT is used as indicator under the "growth of lowest income share" model specification.

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Empirical evidence suggests that Stock market expansion has been correlated by higher income inequality in developed economies. As already argued by Picketty (2014), people with higher incomes are likely to earn higher return on capital invested given their ability to bear more risk. This finding is supported by Owyang and Shell (2016) who argue that co-movement between stock prices and income inequality is explained by the fact that gains in the stock market are likely to concentrated at the top.

To assess whether the findings above are robust, we use data averaged over five year periods. The sample data covers the period 1980-2014. Data are gathered from the same sources as for the cross-sectional study, except for income inequality data which were obtained from the SWIID Database (Solt, 2016). The estimation results are reported in Table 7[7]. The linearity tests clearly indicate that there is a significant financial development threshold, both for stock-market-based and bank-based financial systems. Moreover, similarly to the cross-sectional study, we found no evidence for an inverted between stock market development and income inequality, while this U-shaped relationship is evident for banking sector development.

INSERT TABLE 7 HERE

5. Discussion

In this paper, we used the Instrumental Variable Threshold Regression introduced by Caner and Hansen (2004) to investigate the relationship between financial development and income inequality using a sample of 60 developed and developing countries over the period between 1980 and 2009.

[7] Only results for models using PRIV and VT are reported here. Linearity tests reject the hypothesis of a threshold effect when LLY_0, DBA_0, $MCAP_0$ and TOR_0 are used as potential threshold variables.

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Linearity tests show strong and significant threshold effects in the relationship between financial development and income inequality. This result is robust to different financial development indicators. In particular, financial intermediary and markets development is found to reduce income inequality only if the country has reached a given threshold level of financial development. That is, during its first stages of development, financial intermediary accentuates inequalities and disproportionately helps more the rich. The positive distributional effects of financial development are not effective unless the banking sector has reached a minimum critical size.

The significant threshold effect in the financial development-inequality link implies that the long-run relationship between financial development and income inequality is nonlinear. This finding means that in early stages of financial development financial imperfections arising from asymmetric information and transaction costs produce credit constraints that are particularly restrictive for small businesses and poor people lacking collateral, credit histories, and connections. By reducing financial frictions, and allowing the poorest people and the least connected entrepreneurs to obtain external funding, financial development can improve the allocation of capital towards more productive investments and, thus, reduce income inequality (Greenwood and Jovanovic, 1990; Kim and Lin, 2011).

Our results show no evidence of an inverted U-shaped relationship between income inequality and stock market development indicators. Unlike Kim and Lin (2011), we found that below a given critical level stock market development has no effect on income inequality. This is to say that in such a lower regime, the weight of the market is very light and its contribution to income distribution does not matter. This result can be explained by the effect of financial integration and liberalization of stock markets. The gains from the liberalization of the stock market seem to be in favor of the top income quintile, to the detriment of the middle quintile and without any effects on the bottom (Das and Mohapatra, 2003). In a bank-based financial system, however,

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an inverted U-shaped relationship between the income inequality and bank development is observed, while we report a U-shaped relationship between the growth of the average income of the poorest quintile and the expansion of the banking sector. These findings suggest that the effect of bank development on inequality is clearly larger than for the stock market, indicating the greater importance of the role of banks for income inequality. This finding especially applies to upper-middle and high income countries. Estimation results indicate that for these countries stock market expansion significantly increases income inequality. Empirical evidence already highlighted this concern. As explained by Moldonado (2017), in a market-based financial system, stock markets offer a wider choice and higher returns for people earning high incomes, resulting in a higher income inequality compared to a more bank-based financial system.

Our empirical results show interesting findings regarding some control variables. First, countries with initially less equitable distribution of income might experience faster reductions in income inequality. However, these reductions are expected to be more important in those countries with a more developed banking system. Besides, the coefficient estimates suggest that schooling significantly reduces income inequalities and is pro-poor, regardless of the level of development of the financial system. This result confirms the tight relationship between access to education, economic growth and poverty alleviation. Furthermore, we found that rapidly growing countries might experience faster reductions in income inequality. Nevertheless, these reductions are expected to be more effective with a less developed banking system.

Overall, our findings have important policy implications, especially for low and lower-middle income countries. In these countries, indeed, where the financial system is essentially bank-based, people earning high incomes are likely to benefit more than the poor from credit-financed investment opportunities (Denk and Cazenave-Lacroutz, 2015). The benefits from a banking sector expansion would be limited to the top of the pyramid. Thus, increased access to finance that gives poor and non-poor people new and better growth opportunities should be

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considered as a priority in these countries in order to reduce income inequalities. Microfinance institutions, cooperatives, credit unions and saving banks can play a key role to achieve this goal. But because the cost of microfinance is relatively high, governments are expected to play an determinant role by lowering the cost of access to microfinance through subsidies and encouraging innovation and diffusion of new technologies into the financial system. Governments can also improve financial access by fostering competition between financial intermediaries which leads to better allocational efficiency.

Finally, the channels through which financial development can help reducing income inequalities are yet to be investigated. Besides, it would be interesting to see how the relationship between these two variables is evolving over time, within and between countries. Capturing time variability in addition to cross-sectional heterogeneity would result in a better understanding of the relation between the two variables.

REFERENCES

Aghion, P., & Bolton, P. (1992). Distribution and growth in models of imperfect capital markets. *European Economic Review, 36*, 603-611.

Aghion, P., & Bolton, P. (1997). A theory of trickle-down growth and development. *The Review of Economic Studies, 64*, 151-172.

Altunbaş, Y., & Thornton, J. (2019). The impact of financial development on income inequality: A quantile regression approach. *Economics Letters, 175*, 51-56.

Ang, J. B. (2010). Finance and inequality: the case of India. *Southern economic journal, 76*, 738-761.

Baiardi, D., & Morana, C. (2018). Financial development and income distribution inequality in the euro area. *Economic Modelling, 70*, 40-55.

Banerjee, A. V., & Newman, A. F. (1993). Occupational choice and the process of development. *Journal of political economy, 101*, 274-298.

Beck, T., & Demirguc-Kunt, A. (2009). *Financial institutions and markets across countries and over time-data and analysis*: The World Bank.

Beck, T., Demirguc-Kunt, A., & Levine, R. (2004). Finance, inequality and poverty: Cross-country evidence (World Bank Policy Research Working Paper 3338). *Washington DC: World Bank*.

Beck, T., Demirgüç-Kunt, A., & Levine, R. (2007). Finance, inequality and the poor. *Journal of economic growth, 12*, 27-49.

Electronic copy available at: https://ssrn.com/abstract=3518099

Beck, T., & Levine, R. (2002). Industry growth and capital allocation:: does having a market- or bank-based system matter? *Journal of financial economics, 64*, 147-180.
Beck, T., & Levine, R. (2004). Stock markets, banks, and growth: Panel evidence. *Journal of Banking & Finance, 28*, 423-442.
Beck, T., Levine, R., & Levkov, A. (2007). Big bad banks? The impact of US branch deregulation on income distribution. *NBER working paper*.
Bonfiglioli, A. (2005). Equities and inequality.
Bresson, F. (2004). Effet du développement financier sur les inégalités de revenus au travers des investissements productifs et éducatifs. *CERDI, Faculté des Sciences Economiques de Clermont-Ferrand*.
Canavire-Bacarreza, G. J., & Rioja, F. K. (2009). Financial development and the distribution of income in Latin America and the Caribbean.
Caner, M., & Hansen, B. E. (2004). Instrumental variable estimation of a threshold model. *Econometric Theory, 20*, 813-843.
Chen, W., & Kinkyo, T. (2016). Financial Development and Income Inequality: Long-Run Relationship and Short-Run Heterogeneity. *Emerging Markets Finance & Trade, 52*, 733-742.
Chiu, Y.-B., & Lee, C.-C. (2019). Financial development, income inequality, and country risk. *Journal of International Money and Finance*.
Clarke, G. R., Xu, L. C., & Zou, H.-f. (2006). Finance and income inequality: what do the data tell us? *Southern economic journal*, 578-596.
Das, M., & Mohapatra, S. (2003). Income inequality: the aftermath of stock market liberalization in emerging markets. *Journal of Empirical Finance, 10*, 217-248.
Demirgüç-Kunt, A., & Levine, R. (2004). *Financial structure and economic growth: A cross-country comparison of banks, markets, and development*: MIT press.
Denk, O., & Cazenave-Lacroutz, A. (2015). Household finance and income inequality in the euro area.
Djankov, S., McLiesh, C., & Shleifer, A. (2005). Private Credit in 129 Countries,‖ National Bureau of Economic Research Working Paper No. 11078. In: Cambridge, MA: NBER.
Dollar, D., & Kraay, A. (2002). Growth is Good for the Poor. *Journal of economic growth, 7*, 195-225.
Easterly, W., & Fischer, S. (2001). Inflation and the Poor. *Journal of Money, Credit and Banking*, 160-178.
Galor, O., & Moav, O. (2004). From physical to human capital accumulation: Inequality and the process of development. *The Review of Economic Studies, 71*, 1001-1026.
Galor, O., & Zeira, J. (1993). Income distribution and macroeconomics. *The Review of Economic Studies, 60*, 35-52.
Gimet, C., & Lagoarde-Segot, T. (2011). A closer look at financial development and income distribution. *Journal of Banking & Finance, 35*, 1698-1713.
Greenwood, J., & Jovanovic, B. (1990). Financial development, growth, and the distribution of income. *Journal of political economy, 98*, 1076-1107.
Haber, S., Maurer, N., & Razo, A. (2003). *The politics of property rights: political instability, credible commitments, and economic growth in Mexico, 1876-1929*: Cambridge University Press.
Hansen, B. E. (1996). Inference when a nuisance parameter is not identified under the null hypothesis. *Econometrica: Journal of the econometric society*, 413-430.
Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica, 68*, 575-603.
Hsiao, C. (1986). Analysis of Panel Data, Econometric Society Monograph No. 11. In: Cambridge: Cambridge University Press.

Electronic copy available at: https://ssrn.com/abstract=3518099

Jauch, S., & Watzka, S. (2016). Financial development and income inequality: a panel data approach. *Empirical Economics, 51*, 291-314.
Kim, D.-H., & Lin, S.-C. (2011). Nonlinearity in the financial development–income inequality nexus. *Journal of Comparative Economics, 39*, 310-325.
Kunieda, T., Okada, K., & Shibata, A. (2014). Finance and inequality: How does globalization change their relationship? *Macroeconomic Dynamics, 18*, 1091-1128.
La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. (1999). The quality of government. *The Journal of Law, Economics, and Organization, 15*, 222-279.
La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (1997). Legal determinants of external finance. *The journal of finance, 52*, 1131-1150.
Landes, D. S. (1998). The wealth and poverty of nations: Why are some so rich and others so poor. In: New York: WW Norton.
Liang, Z. (2006). Financial development and income distribution: A system GMM panel analysis with application to urban China. *Journal of economic development, 31*, 1.
Liu, G., Liu, Y., & Zhang, C. (2017). Financial Development, Financial Structure and Income Inequality in China. *The World Economy, 40*, 1890-1917.
Lynch, D. (1996). Measuring financial sector development: a study of selected Asia-Pacific countries. *The Developing Economies, 34*, 1-27.
Maldonado, J. E. (2017). Financial Systems and Income Inequality. . In *Bruges European Economic Research Papers* College of Europe.
Mankiw, N. G., Romer, D., & Weil, D. N. (1992). A contribution to the empirics of economic growth. *The quarterly journal of economics, 107*, 407-437.
Nasreddine, K., & Mensi, S. (2016). Financial Development and Income Inequality: The Linear versus the Nonlinear Hypothesis. *Economics Bulletin, 36*, 609-626.
OECD. (2011). An Overview of Growing Income Inequalities in OECD Countries: Main Findings. In *Divided We Stand: Why Inequality Keeps Rising* (pp. 21-45). Paris: OECD Publishing.
OECD(a). (2015). *In it together: Why less inequality benefits all*: OECD publishing.
OECD(b). (2015). How to restore a healthy financial sector that supports long-lasting, inclusive growth? In *OECD Economics Department Policy Notes*.
Ostry, J. D., & Berg, A. (2011). Inequality and Unsustainable Growth; Two Sides of the Same Coin? In: International Monetary Fund.
Owyang, M., & Shell, H. (2016). Taking Stock: Income Inequality and the Stock Market.
Park, D., & Shin, K. (2017). Economic growth, financial development, and income inequality. *Emerging Markets Finance and Trade, 53*, 2794-2825.
Pesaran, M. H., & Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of econometrics, 68*, 79-113.
Piketty, T. (1997). The dynamics of the wealth distribution and the interest rate with credit rationing. *The Review of Economic Studies, 64*, 173-189.
Piketty, T. (2014). *Capital in the 21st Century*: The Belknap Press of Harvard University Press.
Porta, R. L., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (1998). Law and finance. *Journal of political economy, 106*, 1113-1155.
Rousseau, P. L., & Wachtel, P. (2000). Equity markets and growth: cross-country evidence on timing and outcomes, 1980–1995. *Journal of Banking & Finance, 24*, 1933-1957.
Sehrawat, M., & Giri, A. (2015). Financial development and income inequality in India: an application of ARDL approach. *International Journal of Social Economics, 42*, 64-81.
Seven, U., & Coskun, Y. (2016). Does financial development reduce income inequality and poverty? Evidence from emerging countries. *Emerging Markets Review, 26*, 34-63.

Electronic copy available at: https://ssrn.com/abstract=3518099

Shahbaz, M., Bhattacharya, M., & Mahalik, M. K. (2017). Finance and income inequality in Kazakhstan: evidence since transition with policy suggestions. *Applied Economics*, *49*, 5337-5351.

Solt, F. (2016). The standardized world income inequality database. *Social science quarterly*, *97*, 1267-1281.

Stand, D. W., & Rising, W. I. K. (2011). An overview of growing income inequalities in OECD countries: main findings. *Divided We Stand: Why Inequality Keeps Rising*.

Stiglitz, J. E. (2015). The origins of inequality, and policies to contain it. *National Tax Journal*, *68*, 425.

Tan, H.-B., & Law, S.-H. (2012). Nonlinear dynamics of the finance-inequality nexus in developing countries. *The Journal of Economic Inequality*, *10*, 551-563.

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Table A1. ***Data sources and definitions***

Variable	Definition	Source
Growth rate of the Gini Coefficient (GINI_GR)	The annual growth rate of the Gini index. It is calculated as the log difference between the last and the first observation divided by the number of years between these two observations.	World Bank Database, WDI
Growth of Lowest Income Share (LOWEST_GR)	The annual growth rate of the income share of the poorest quintile compared to the total national income. It is calculated as the log difference between the last and the first observation divided by the number of years between these two observations.	World Bank Database, WDI
Private Credit (PRIV)	All funds distributed by domestic commercial banks and financial institutions to the private sector, measured by the ratio of private credit / GDP.	Financial Structure dataset (Beck et al (2009)
Liquid Liability Ratio (LLY)	This measure is also known as the liquidity ratio, it is measured by the ratio of M2 to GDP.	Financial Structure dataset (Beck et al (2009)
Deposit money banks assets (DBA)	Total assets held by domestic commercial banks relative to GDP.	Financial Structure dataset (Beck et al (2009)
Turnover Ratio (TOR)	The value of traded shares on domestic exchanges divided by the value of listed shares.	Financial Structure dataset (Beck et al (2009)
Value traded (VT)	The value of traded shares on domestic exchanges divided by GDP	Financial Structure dataset (Beck et al (2009)
Stock Market Capitalization (MCAP)	Total value of listed shares divided by GDP. It indicates the size of stock market relative to the size of the economy	Financial Structure dataset (Beck et al (2009)
Growth rate of real GDP per capita (GDP_GR)	Annual growth rate of real GDP per capita (GDP in constant international dollars).	World Bank Database, WDI
Secondary school attainment (SCHOOL)	The average number of school years (in 1980)	Barro R. & J.W. Lee (2010)
Inflation (INFL)	The inflation rate represented by the increase in consumption price index for a period of one year	World Bank Database, WDI
Government expenditure (GOV)	The value of final consumption expenditure of government relative to GDP.	World Bank Database, WDI
Trade openness (TRADE)	The ratio (exports + imports) / GDP.	World Bank Database, WDI
Creditor rights(CREDITOR)	The legal rights of creditors against defaulters. The index varies from 1 to 4. A higher value indicates stronger creditor rights.	Djankov et al (2005)
legal origin (BRITISH)	A dummy variable taking the value 1 or 0. It indicate whether the legal system (legal) tradition is based on the English or French	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1999)
Religious composition (MUSLIM, CATHOLIC)	The fraction of the population which is Muslim, Catholic, Protestant or other religions. (in% of the population in 1980)	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1999)
Ethnic fractionalization (ETHNIC)	The probability that two randomly selected individuals from a country are from different ethnolinguistic groups.	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1999)

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Table A.2. ***IV Linear regression-GMM***

Dependent variable	*GINI_GR*						*LOWEST_GR*					
Variable	*PRIV*	*LLY*	*DBA*	*MCAP*	*TOR*	*VT*	*PRIV*	*LLY*	*DBA*	*MCAP*	*TOR*	*VT*
Financial development variable coefficient	-0.112	-0.145	0.208	-0.020	-0.098	-0.126	-0.445	0.658	-0.154	0.373	0.376	0.165
	(0.6169)	(0.6489)	(0.4107)	(0.8318)	(0.3253)	(0.2682)	(0.2480)	(0.2167)	(0.7191)	(0.2438)	(0.1317)	(0.1888)
C	3.877*	6.152***	2.543	3.422***	4.797***	5.782***	2.303	0.665	2.471	-0.197	0.186	1.540
	(0.0574)	(0.0023)	(0.1983)	(0.0177)	(0.0048)	(0.0015)	(0.2726)	(0.7081)	(0.1175)	(0.8918)	(0.9043)	(0.2226)
$GINI_0$	-1.086**	-1.394***	-0.772**	-0.769**	-1.086***	-1.39***						
	(0.0125)	(0.0006)	(0.0413)	(0.0211)	(0.0008)	(0.0010)						
$LOWEST_0$							-1.458***	-1.676***	-1.562***	-1.677***	-2.169***	-2.001***
							(0.0025)	(0.0029)	(0.0018)	(0.0000)	(0.0001)	(0.0000)
GDP_GR_0	0.039*	-0.009	0.027*	0.053***	0.041***	0.012	-0.019	0.033	0.010	-0.029	-0.053*	-0.042*
	(0.0631)	(0.6942)	(0.0565)	(0.0000)	(0.0004)	(0.4939)	(0.5529)	(0.3664)	(0.7737)	(0.3605)	(0.0513)	(0.0799)
$SCHOOL_0$	-0.200	-0.085	-0.352**	-0.320*	-0.261*	-0.050	0.499	-0.088	0.446*	-0.512	-0.058	-0.093
	(0.1954)	(0.6046)	(0.0133)	(0.0806)	(0.0853)	(0.8424)	(0.2128)	(0.7426)	(0.0521)	(0.3065)	(0.8673)	(0.7973)
$TRADE_0$	0.403**	0.461***	0.293*	0.259***	0.263**	0.332**	-0.377	-0.815***	-0.597*	-0.305	-0.144	-0.254
	(0.0231)	(0.0059)	(0.0710)	(0.0025)	(0.0123)	(0.0108)	(0.2443)	(0.0028)	(0.0649)	(0.1189)	(0.6096)	(0.1150)
GOV0	-0.505*	-0.907***	-0.619***	-0.547***	-0.528***	-0.692***	1.407***	1.228***	1.322***	1.469***	1.202***	1.132***
	(0.0741)	(0.0061)	(0.0014)	(0.0000)	(0.0001)	(0.0000)	(0.0017)	(0.0058)	(0.0025)	(0.0003)	(0.0013)	(0.0004)
INFL0	0.01***	0.002	0.005	-0.003***	-0.003**	0.000	-0.008	-0.000	-0.010	-0.005	0.002	-0.001
	(0.0122)	(0.4619)	(0.1816)	(0.0041)	(0.0325)	(0.8908)	(0.3926)	(0.9783)	(0.1735)	(0.1600)	(0.6494)	(0.7265)
J-statistic	7.0361	8.1755	6.3052	6.9651	7.1711	9.0798	6.7175	4.3215	4.5807	5.2828	5.8479	5.1016
Prob(J-stat)	0.2179	0.1468	0.2776	0.2232	0.2082	0.1059	0.2425	0.5041	0.4691	0.3823	0.3213	0.4036
Observations	51	51	51	50	51	51	51	51	51	50	51	51

*, **, *** indicate statistical significance, respectively, at 10%, 5% and 1% level. The p-values are in parentheses. The null hypothesis of Hansen's J-test Hansen (J-statistic) indicates that the instruments are not correlated to the error term, so that the instruments are valid. All variables are in logarithm, except for the inflation rate.

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Table A3-1 : List of countries

Argentina	Greece	Niger
Australia	Guatemala	Norway
Austria	Hondouras	Pakistan
Bangladesh	Hong Kong	Panama
Belgium	Hungary	Peru
Bolivia	India	Philippines
Brazil	Indonesia	Portugal
Cameroon	Iran, Islamic Rep	Senegal
Canada	Ireland	Spain
Chile	Italy	Sri Lanka
Colombia	Jamaica	Sweden
Costa rika	Japan	Tanzania
Denmark	Jordan	Thailand
Dominican Rep	Korea, Rep	Tunisia
Ecuador	Lesotho	Turkey
Egypt, Arab Rep	Malaysia	Uganda
El Salvador	Mexico	United States
Finland	Nepal	Uruguay
France	Netherlands	Venezuela, RB
Ghana	New Zealand	Zambia

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Table A3-2 : Countries classification by financial regime, based on estimations from table 3

Threshold Variables					
$PRIV_0$		LLY_0		DBA_0	
Lower Regime	**Higher Regime**	**Lower Regime**	**Higher Regime**	**Lower Regime**	**Higher Regime**
Argentina	Austria	Argentina	Australia	Argentina	Australia
Australia	Belgium	Bolivia	Austria	Bangladesh	Austria
Bangladesh	Brazil	Brazil	Bangladesh	Bolivia	Belgium
Bolivia	Canada	Cameroon	Belgium	Cameroon	Brazil
Cameroon	Chile	Chile	Canada	Ecuador	Canada
Colombia	El Salvador	Colombia	Costa rika	Ghana	Chile
Costa rika	Finland	Ecuador	Denmark	Guatemala	Colombia
Denmark	France	Ghana	Dominican Rep	Hungary	Costa rika
Dominican Rep	Greece	Guatemala	Egypt, Arab Rep	Indonesia	Denmark
Ecuador	Hondouras	Indonesia	El Salvador	Mexico	Dominican Rep
Egypt, Arab Rep	Hong Kong	Niger	Finland	Nepal	Egypt, Arab Rep
Ghana	Iran, Islamic Rep	Peru	France	New Zealand	El Salvador
Guatemala	Ireland	Philippines	Greece	Niger	Finland
Hungary	Italy	Senegal	Hondouras	Peru	France
India	Jamaica	Tanzania	Hong Kong	Tanzania	Greece
Indonesia	Japan	Turkey	Hungary	Turkey	Hondouras
Lesotho	Jordan	Uganda	India	Uganda	Hong Kong
Mexico	Korea, Rep	Zambia	Iran, Islamic Rep	Zambia	India
Nepal	Malaysia		Ireland		Iran, Islamic Rep
Niger	Netherlands		Italy		Ireland
Pakistan	New Zealand		Jamaica		Italy
Peru	Norway		Japan		Jamaica
Senegal	Panama		Jordan		Japan
Sri Lanka	Philippines		Korea, Rep		Jordan
Tanzania	Portugal		Lesotho		Korea, Rep
Turkey	Spain		Malaysia		Lesotho
Uganda	Sweden		Mexico		Malaysia
Zambia	Thailand		Nepal		Netherlands
	Tunisia		Netherlands		Norway
	United States		New Zealand		Pakistan
	Uruguay		Norway		Panama
	Venezuela, RB		Pakistan		Philippines
			Panama		Portugal
			Portugal		Senegal
			Spain		Spain
			Sri Lanka		Sri Lanka
			Sweden		Sweden
			Thailand		Thailand
			Tunisia		Tunisia
			United States		United States
			Uruguay		Uruguay
			Venezuela, RB		Venezuela, RB

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Table A3-3 : Countries classification by financial regime, based on estimations from table 4

Threshold Variables					
PRIV0		**LLY0**		**DBA0**	
Lower Regime	Higher Regime	Lower Regime	Higher Regime	Lower Regime	Higher Regime
Argentina	Austria	Argentina	Australia	Argentina	Australia
Australia	Canada	Bolivia	Austria	Bangladesh	Austria
Bangladesh	Finland	Brazil	Bangladesh	Bolivia	Belgium
Belgium	France	Cameroon	Belgium	Cameroon	Brazil
Bolivia	Hong Kong	Chile	Canada	Colombia	Canada
Brazil	Ireland	Colombia	Costa rika	Ecuador	Chile
Cameroon	Italy	Ecuador	Denmark	Ghana	Costa rika
Chile	Japan	Ghana	Dominican Rep	Guatemala	Denmark
Colombia	Jordan	Guatemala	Egypt, Arab Rep	Hungary	Dominican Rep
Costa rika	Korea, Rep	Indonesia	El Salvador	Indonesia	Egypt, Arab Rep
Denmark	Malaysia	Niger	Finland	Mexico	El Salvador
Dominican Rep	Netherlands	Peru	France	Nepal	Finland
Ecuador	Norway	Philippines	Greece	New Zealand	France
Egypt, Arab Rep	Panama	Senegal	Hondouras	Niger	Greece
El Salvador	Philippines	Tanzania	Hong Kong	Peru	Hondouras
Ghana	Portugal	Turkey	Hungary	Senegal	Hong Kong
Greece	Spain	Uganda	India	Sri Lanka	India
Guatemala	Sweden	Zambia	Iran, Islamic Rep	Tanzania	Iran, Islamic Rep
Hondouras	Tunisia		Ireland	Turkey	Ireland
Hungary	United States		Italy	Uganda	Italy
India	Venezuela, RB		Jamaica	Zambia	Jamaica
Indonesia			Japan		Japan
Iran, Islamic Rep			Jordan		Jordan
Jamaica			Korea, Rep		Korea, Rep
Lesotho			Lesotho		Lesotho
Mexico			Malaysia		Malaysia
Nepal			Mexico		Netherlands
New Zealand			Nepal		Norway
Niger			Netherlands		Pakistan
Pakistan			New Zealand		Panama
Peru			Norway		Philippines
Senegal			Pakistan		Portugal
Sri Lanka			Panama		Spain
Tanzania			Portugal		Sweden
Thailand			Spain		Thailand
Turkey			Sri Lanka		Tunisia
Uganda			Sweden		United States
Uruguay			Thailand		Uruguay
Zambia			Tunisia		Venezuela, RB
			United States		
			Uruguay		
			Venezuela, RB		

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Table A3-4 : Countries classification by financial regime, based on estimations from table 5

Threshold Variables					
$MCAP_0$		TOR_0		VT_0	
Lower Regime	**Higher Regime**	**Lower Regime**	**Higher Regime**	**Lower Regime**	**Higher Regime**
Argentina	Australia	Bangladesh	Argentina	Argentina	Australia
Bangladesh	Austria	Belgium	Australia	Austria	Belgium
Bolivia	Belgium	Bolivia	Austria	Bangladesh	Canada
Brazil	Canada	Chile	Brazil	Bolivia	Denmark
Costa rika	Chile	Colombia	Canada	Brazil	Finland
Ecuador	Colombia	Costa rika	Denmark	Chile	France
Egypt, Arab Rep	Denmark	Ecuador	Finland	Colombia	Hong Kong
El Salvador	Finland	Egypt, Arab Rep	France	Costa rika	Ireland
Ghana	France	El Salvador	Hong Kong	Ecuador	Italy
Greece	Hong Kong	Ghana	Hungary	Egypt, Arab Rep	Japan
Guatemala	Iran, Islamic Rep	Greece	India	El Salvador	Jordan
Hungary	Ireland	Guatemala	Ireland	Ghana	Korea, Rep
India	Italy	Indonesia	Italy	Greece	Malaysia
Indonesia	Jamaica	Iran, Islamic Rep	Japan	Guatemala	Netherlands
Pakistan	Japan	Jamaica	Jordan	Hungary	New Zealand
Panama	Jordan	Nepal	Korea, Rep	India	Norway
Peru	Korea, Rep	New Zealand	Malaysia	Indonesia	Spain
Portugal	Malaysia	Pakistan	Mexico	Iran, Islamic Rep	Sweden
Tanzania	Mexico	Panama	Netherlands	Jamaica	United States
Thailand	Nepal	Peru	Norway	Mexico	
Turkey	Netherlands	Philippines	Spain	Nepal	
Uganda	New Zealand	Portugal	Sweden	Pakistan	
Uruguay	Norway	Sri Lanka	United States	Panama	
Venezuela, RB	Philippines	Tanzania		Peru	
Zambia	Spain	Thailand		Philippines	
	Sri Lanka	Tunisia		Portugal	
	Sweden	Turkey		Sri Lanka	
	Tunisia	Uganda		Tanzania	
	United States	Uruguay		Thailand	
		Venezuela, RB		Tunisia	
		Zambia		Turkey	
				Uganda	
				Uruguay	
				Venezuela, RB	
				Zambia	

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Table A3-5 : Countries classification by financial regime, based on estimations from table 6

Threshold Variables					
$MCAP_0$		TOR_0		VT_0	
Lower Regime	**Higher Regime**	**Lower Regime**	**Higher Regime**	**Lower Regime**	**Higher Regime**
Argentina	Australia	Argentina	Australia	Argentina	Australia
Bangladesh	Austria	Bangladesh	Austria	Austria	Belgium
Bolivia	Belgium	Belgium	Brazil	Bangladesh	Canada
Brazil	Canada	Bolivia	Canada	Bolivia	Denmark
Colombia	Chile	Chile	Finland	Brazil	Finland
Costa rika	Denmark	Colombia	France	Chile	France
Ecuador	Finland	Costa rika	Hong Kong	Colombia	Hong Kong
Egypt, Arab Rep	France	Denmark	Hungary	Costa rika	Ireland
El Salvador	Hong Kong	Ecuador	India	Ecuador	Italy
Ghana	Iran, Islamic Rep	Egypt, Arab Rep	Ireland	Egypt, Arab Rep	Japan
Greece	Ireland	El Salvador	Italy	El Salvador	Jordan
Guatemala	Italy	Ghana	Japan	Ghana	Korea, Rep
Hungary	Jamaica	Greece	Korea, Rep	Greece	Malaysia
India	Japan	Guatemala	Mexico	Guatemala	Netherlands
Indonesia	Jordan	Indonesia	Netherlands	Hungary	New Zealand
Mexico	Korea, Rep	Iran, Islamic Rep	Norway	India	Norway
Nepal	Malaysia	Jamaica	Spain	Indonesia	Spain
Pakistan	Netherlands	Jordan	United States	Iran, Islamic Rep	Sweden
Panama	New Zealand	Malaysia		Jamaica	United States
Peru	Norway	Nepal		Mexico	
Portugal	Philippines	New Zealand		Nepal	
Sri Lanka	Spain	Pakistan		Pakistan	
Tanzania	Sweden	Panama		Panama	
Thailand	United States	Peru		Peru	
Tunisia		Philippines		Philippines	
Turkey		Portugal		Portugal	
Uganda		Sri Lanka		Sri Lanka	
Uruguay		Sweden		Tanzania	
Venezuela, RB		Tanzania		Thailand	
Zambia		Thailand		Tunisia	
		Tunisia		Turkey	
		Turkey		Uganda	
		Uganda		Uruguay	
		Uruguay		Venezuela, RB	
		Venezuela, RB		Zambia	
		Zambia			

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Table 1. ***Descriptive statistics***

	GINI_GR	*LOWEST_GR*	*PRIV*	*LLY*	*DBA*	*MCAP*	*TOR*	*VT*
Mean	1.309673	4.948605	46.84898	49.83539	52.04848	35.16637	33.6942	16.89418
Median	0.94479	1.348356	34.08286	44.0246	39.95653	22.60716	24.68379	8.8747
Standard Deviation	1.549268	11.78227	34.27407	30.7831	36.04627	34.86344	30.973	21.36475
Minimum	0.099438	0.012287	4.973411	12.51126	7.384201	0.823503	0.708862	0.01946
Maximum	10.05651	60.7757	146.624	169.1166	205.3176	173.1684	125.9846	95.15751
Obs.	60	60	60	60	60	54	54	54

Table 2.1. ***Correlation matrix***

	GINI_GR	**LOWEST_GR**	**PRIV**	**LLY**	**DBA**	**MCAP**	**TOR**	**VT**
GINI_GR	1.000000							
LOWEST_GR	-0.800862	1.000000						
PRIV	-0.242666	0.224567	1.000000					
LLY	-0.103196	0.108795	0.864547	1.000000				
DBA	-0.175000	0.167320	0.919517	0.931271	1.000000			
MCAP	-0.190609	0.180387	0.662594	0.626633	0.658674	1.000000		
TOR	-0.324871	0.290732	0.521461	0.462854	0.536523	0.569943	1.000000	
VT	-0.286483	0.262035	0.677955	0.635380	0.686828	0.865693	0.895168	1.000000

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Table 2.2. Randomness tests

	lpriv0	llly0	ldba0	lmcap0	ltor	lvt0
h	0	0	0	0	0	0
p	0.1479	0.5913	0.1479	0.6540	0.9806	0.5148

H is the hypothesis test result, p is the p-value. The test returns a test decision for the null hypothesis that the values in the data vector x come in random order, against the alternative that they do not. The test is based on the number of runs of consecutive values above or below the mean of x. The result h is 1 if the test rejects the null hypothesis at the 5% significance level, or 0 otherwise. Results are obtained by running the Wald–Wolfowitz "Runstest" under Matlab.

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Table 3. ***IV Threshold effects of Financial Intermediaries development on Growth of Gini***

	Dependent variable: GINI_GR					
THRESHOLD VARIABLE	**$PRIV_0$**		**LLY_0**		**DBA_0**	
Sup W	161.5936**		109.6451*		137.5556*	
	(0.0109)		(0.0672)		(0.0708)	
ESTIMATED THRESHOLD ($\hat{\gamma}$)	***3.2744***		***3.2227***		***3.2746***	
	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$
Financial development variable coefficient	***1.2060****	***-0.6190****	***2.7506*****	***-0.6139****	***2.2352*****	***-0.2368***
	(0.0787)	**(0.0630)**	**(0.0020)**	**(0.0530)**	**(0.0014)**	**(0.2906)**
C	3.5107	6.9763**	-4.7770	6.3842***	-1.0946	3.3215**
	(0.1650)	(0.0270)	(0.1959)	(0.0015)	(0.6359)	(0.0416)
$GINI_0$	-1.5378***	-1.6284***	-0.5857	-1.4220***	-1.2339***	-0.6254*
	(0.0008)	(0.0061)	(0.4806)	(0.0005)	(0.0033)	(0.0704)
GDP_GR	-0.0923**	0.1292***	-0.0908**	0.0614***	-0.0917***	0.0665***
	(0.0388)	(0.0006)	(0.0143)	(0.0012)	(0.0039)	(0.0008)
SCHOOL	-0.7412*	-0.6197**	-0.5420**	-0.6012***	-0.8022***	-0.4077**
	(0.0563)	(0.0229)	(0.0413)	(0.0008)	(0.0030)	(0.0112)
TRADE	0.3292	0.1487	-0.4510	0.3333***	-0.1590	0.4175***
	(0.1367)	(0.4303)	(0.3203)	(0.0088)	(0.5358)	(0.0052)
GOV	-1.2791***	0.2451	-0.2805	0.0026	-0.4197**	-0.6910***
	(0.0064)	(0.6687)	(0.6217)	(0.9946)	(0.0211)	(0.0017)
INFL	0.0177**	0.0136**	0.0094***	0.0002	0.0170***	0.0086**
	(0.0458)	(0.0278)	(0.0076)	(0.9807)	(0.0025)	(0.0320)
Obs.	23	28	16	35	16	35

*, **, *** indicate statistical significance, respectively, at 10%, 5% and 1% level. The p-values are in parentheses.
All variables are in logarithm, except for the inflation rate. As threshold variables, we consider the initial values of PRIV, LLY and DBA.

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Table 4. ***IV Threshold effects of Financial Intermediaries development on Growth of Lowest Income Share***

	Dependent variable :LOWEST_GR					
THRESHOLD VARIABLE	**$PRIV_0$**		**LLY_0**		**DBA_0**	
Sup W	140.9805**		103.9821*		357.2119***	
	(0.0451)		(0.0678)		(0.0015)	
ESTIMATED THRESHOLD ($\hat{\gamma}$)	***3.6937***		***3.2227***		***3.2924***	
	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$
Financial development variable coefficient	***-1.3209******	***1.9448****	***-5.1069******	***1.1895****	***-3.8977******	***0.1768***
	(0.0007)	**(0.0490)**	**(0.0078)**	**(0.0531)**	**(0.0002)**	**(0.7286)**
C	5.4359**	8.4686**	11.2901***	3.8156*	7.1121*	4.5289***
	(0.0195)	(0.0430)	(0.0065)	(0.0859)	(0.0664)	(0.0005)
$LOWEST_0$	-1.1673*	-7.4914***	-0.6407	-2.3252***	-2.7053***	-1.9614***
	(0.0868)	(0.0003)	(0.4053)	(0.0000)	(0.0000)	(0.0007)
GDP_GR	0.0224	-0.4726***	0.1773**	-0.0532*	0.3256***	-0.0786
	(0.5837)	(0.0004)	(0.0494)	(0.0535)	(0.0000)	(0.1037)
SCHOOL	0.6468	0.8541	1.0250**	0.7932***	1.1072***	0.7896***
	(0.1058)	(0.2050)	(0.0392)	(0.0022)	(0.0050)	(0.0029)
TRADE	-1.0451***	-0.6506	0.7332	-0.8186***	0.7733	-0.6077**
	(0.0012)	(0.1410)	(0.3330)	(0.0001)	(0.2471)	(0.0402)
GOV	2.0978***	-0.0271	1.7238	-0.2951	2.6394***	0.3286
	(0.0028)	(0.9553)	(0.1072)	(0.4850)	(0.0020)	(0.4743)
INFL	-0.0083	0.0272	-0.0137**	-0.0118	0.0030	-0.0180**
	(0.3681)	(0.3451)	(0.0384)	(0.5846)	(0.8120)	(0.0449)
Obs.	33	18	16	35	16	35

*, **, *** indicate statistical significance, respectively, at 10%, 5% and 1% level. The p-values are in parentheses.
All variables are in logarithm, except for the inflation rate. As threshold variables, we consider the initial values of PRIV, LLY and DBA.

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Table 5. ***IV Threshold effects of Stock Market development on Growth of Gini***

	Dependent variable : GINI_GR					
THRESHOLD VARIABLE	**$MCAP_0$**		**TOR_0**		**VT_0**	
Sup W	93.8177*		113.5371*		164.8736**	
	(0.0733)		(0.0531)		(0.0325)	
ESTIMATED THRESHOLD ($\hat{\gamma}$)	***1.5833***		***2.7645***		***0.9440***	
	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$
Financial development variable coefficient	***0.1255***	***0.3395****	***-0.1026***	***0.9618*****	***-0.0095***	***-0.1256***
	(0.2948)	**(0.0303)**	**(0.4252)**	**(0.0080)**	**(0.9275)**	**(0.3101)**
C	4.2516***	6.3952**	3.8450**	-0.4057	5.0082***	-2.2270
	(0.0023)	(0.0185)	(0.0430)	(0.8299)	(0.0046)	(0.5588)
$GINI_0$	-1.1890***	-1.5045**	-1.1556***	-1.3656***	-1.1829***	0.5481
	(0.0031)	(0.0186)	(0.0023)	(0.0001)	(0.0040)	(0.4444)
GDP_GR	-0.0019	0.0637***	0.0237	0.0926***	0.0095	-0.2662**
	(0.9258)	(0.0000)	(0.2436)	(0.0001)	(0.6894)	(0.0207)
SCHOOL	-0.1774	-0.9979***	-0.1670	-0.5207***	-0.1772	-0.0862
	(0.2334)	(0.0004)	(0.2328)	(0.0004)	(0.3468)	(0.6306)
TRADE	0.1660	-0.0072	0.3919**	0.6625***	0.1150	1.1349***
	(0.3086)	(0.9507)	(0.0392)	(0.0006)	(0.5620)	(0.0027)
GOV	-0.2040	-0.6058***	-0.2522	-0.4554***	-0.3439	-1.3942***
	(0.4447)	(0.0060)	(0.1942)	(0.0021)	(0.1892)	(0.0017)
INFL	-0.0019	-0.0005	-0.0037*	0.0147***	-0.0005	0.0128
	(0.3050)	(0.9655)	(0.0720)	(0.0036)	(0.8349)	(0.5643)
Obs.	23	27	31	20	35	16

*, **, *** indicate statistical significance, respectively, at 10%, 5% and 1% level. The p-values are in parentheses.
All variables are in logarithm, except for the inflation rate. As threshold variables, we consider the initial values of MCAP, TOR and VT.

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Table 6. ***IV Threshold effects of Stock Market development on Growth of Lowest Income Share***

	Dependent variable : LOWEST_GR					
THRESHOLD VARIABLE	**$MCAP_0$**		**TOR_0**		**VT_0**	
Sup W	103.0779*		144.6765*		430.7137***	
	(0.0765)		(0.0617)		(0.0009)	
ESTIMATED THRESHOLD ($\hat{\gamma}$)	***1.8717***		***3.0328***		***0.8157***	
	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$
Financial development variable coefficient	**-0.3419**	**-0.1952**	**0.0744**	**1.2387****	**-0.0631**	**-0.5314****
	(0.4661)	**(0.6263)**	**(0.8149)**	**(0.0431)**	**(0.7812)**	**(0.0138)**
C	1.5960	2.3640	1.5306	7.1436	1.1691	11.1271***
	(0.2606)	(0.2480)	(0.4594)	(0.2045)	(0.4898)	(0.0004)
$LOWEST_0$	-1.4956***	-2.3027***	-1.6900***	-5.6886***	-1.5984***	-4.4384***
	(0.0004)	(0.0089)	(0.0045)	(0.0000)	(0.0015)	(0.0000)
GDP_GR	0.0150	-0.1247**	0.0003	-0.1824***	0.0259	-0.2052***
	(0.8099)	(0.0268)	(0.9959)	(0.0020)	(0.6265)	(0.0001)
SCHOOL	-0.7437	1.4133***	-0.0235	-0.3940	-0.0206	0.5419
	(0.1446)	(0.0004)	(0.9223)	(0.1910)	(0.9624)	(0.1471)
TRADE	-0.4571*	0.0857	-0.6738	0.8515***	-0.6157**	-0.3131**
	(0.0909)	(0.8335)	(0.1370)	(0.0036)	(0.0264)	(0.0213)
GOV	1.5828**	0.6253	1.4993***	-0.8690	1.6036**	0.0219
	(0.0300)	(0.1182)	(0.0022)	(0.3192)	(0.0162)	(0.9353)
INFL	-0.0063**	-0.0053	-0.0003	-0.0717***	-0.0033	0.0557***
	(0.0169)	(0.8203)	(0.9588)	(0.0008)	(0.5234)	(0.0062)
Obs.	27	23	35	16	34	17

*, **, *** indicate statistical significance, respectively, at 10%, 5% and 1% level. The p-values are in parentheses.
All variables are in logarithm, except for the inflation rate. As threshold variables, we consider the initial values of MCAP, TOP and VT.

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Table 7. ***IV Threshold effects of Financial Intermediaries development on Growth of Gini-5 years-average***

THRESHOLD VARIABLE	**$PRIV_0$**		**VT_0**	
Sup W	35.9460*		48.2822***	
	(0.07)		(0.0185)	
ESTIMATED THRESHOLD ($\hat{\gamma}$)	***4.414***		***0.6378***	
	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$	$q_i \leq \hat{\gamma}$	$q_i > \hat{\gamma}$
Financial development variable coefficient	***0.2233***	***-2.2763****	***-0.8149****	***-0.3382****
	(0.3850)	**(0.0149)**	**(0.0127)**	**(0.0031)**
C	3.2307	34.8078***	11.9393	9.3803***
	(0.1853)	(0.0000)	(0.1510)	(0.0003)
$GINI_0$	-0.2256	-5.3156***	-2.8883	-1.5070***
	(0.6967)	(0.0001)	(0.1651)	(0.0249)
GDP_GR	-0.0043	0.2872**	-0.4111*	0.1147***
	(0.9444)	(0.0223)	(0.1085)	(0.0610)
SCHOOL	0.7996***	-0.4328	1.5459***	1.3115***
	(0.0036)	(0.3592)	(0.0073)	(0.0002)
TRADE	-0.3654	-0.7479***	0.0328	-0.5520***
	(0.1529)	(0.0029)	(0.9635)	(0.0047)
GOV	0.0798	0.6203*	0.3386	0.0525
	(0.5744)	(0.0591)	(0.4794)	(0.7548)
Obs.	234	70	46	214

*, **, *** indicate statistical significance, respectively, at 10%, 5% and 1% level. The p-values are in parentheses. All variables are in logarithm. As threshold variables, we consider the initial values of PRIV and VT.

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